Task 13: Configuration Management using Puppet

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1. Objective

Use Puppet to define and maintain the desired configuration of the Delivery Promise Calculator deployment environment.

2. Puppet Installation

```
puppet --version
# 8.10.0

puppet config print vardir
# C:/Users/vedan/.puppetlabs/opt/puppet/cache
```

3. Puppet Manifest

```
# Task 13 - Delivery Promise Calculator
# Configuration management using Puppet

$delivery_root = 'C:/DeliveryPromise'

file { $delivery_root:
  ensure => directory,
}

file { "${delivery_root}/config":
  ensure  => directory,
  require => File[$delivery_root],
}

file { "${delivery_root}/logs":
  ensure  => directory,
  require => File[$delivery_root],
}

file { "${delivery_root}/config/application.properties":
  ensure  => file,
  content => @("PROPERTIES"),
spring.application.name=delivery-promise-calculator
server.port=8765
| PROPERTIES
  require => File["${delivery_root}/config"],
}
```

4. Apply and Verify

```
puppet apply puppet\delivery_promise.pp
Get-ChildItem C:\DeliveryPromise
Get-Content C:\DeliveryPromise\config\application.properties
```

5. Initial Result

Puppet created the config and logs directories and defined the application.properties content.

6. Idempotency Test

Applying the same manifest a second time produced only catalog compilation/application notices and no resource creation/change messages.

7. Git Integration

```
git checkout -b feature/puppet-configuration
git add puppet/delivery_promise.pp
git commit -m "Add Puppet configuration management"
git push -u origin feature/puppet-configuration
# PR -> main -> merge
```

8. Result

Puppet successfully established a repeatable desired state and the manifest was merged into main.

Task 13 — Actual Project Screenshots

Screenshots captured during the implementation are included below as evidence. These are the actual project screenshots, not recreated diagrams.

Figure 1. Puppet manifest application and verification of the managed application.properties file.

